



COURSE

ADVANCED DATA SCIENCE LAB

CODE

MCA33PE21

FORMATIVE ASSESSMENT

01

Advanced Data Science Lab [MCA33PE21]

Field	Details
Academic Year	2026-2027
Class / Division	SYMCA (Semester-I)
PRN	125M1H064
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Course Teacher	Prof. Prakash Ukhalkar
Submission Date	27th of August 2026

Topic: Coloured Pairplot

- Explain pairplots with hue.
- Generate pairplots on datasets with category variable.
- Interpret how categories are separated in feature space.
- Explain each step briefly.

Objective

The objective of this activity is to understand and generate a coloured pairplot using the Iris dataset. The `hue` parameter is used to distinguish different species using different colours. The pairplot helps visualize relationships between numerical features and understand how different species are separated in feature space.

Importing the Libraries.

```
In [1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

Using Sklearn's Iris dataset.

Explanation

The Iris dataset is a built-in dataset provided by Scikit-learn. It contains four numerical features:

- Sepal length
- Sepal width
- Petal length
- Petal width

It also contains three flower species: Setosa, Versicolor, and Virginica.

```
In [2]: from sklearn.datasets import load_iris

# Loading Iris dataset
iris = load_iris()
```

```
In [19]: # Creating DataFrame
iris_df['species'] = iris.target
iris_df['species_name'] = iris_df['species'].apply(lambda x: iris.target_names[x])

iris_df
```

Out[19]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species	species_name
0	5.1	3.5	1.4	0.2	0	setosa
1	4.9	3.0	1.4	0.2	0	setosa
2	4.7	3.2	1.3	0.2	0	setosa
3	4.6	3.1	1.5	0.2	0	setosa
4	5.0	3.6	1.4	0.2	0	setosa
...	...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	2	virginica
146	6.3	2.5	5.0	1.9	2	virginica
147	6.5	3.0	5.2	2.0	2	virginica
148	6.2	3.4	5.4	2.3	2	virginica
149	5.9	3.0	5.1	1.8	2	virginica

150 rows × 6 columns

```
In [12]: # Displaying first rows
print("First 3 classes of the dataset:")
print(iris_df.iloc[[1,50,100], :])
```

First 3 classes of the dataset:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
1	4.9	3.0	1.4	0.2	
50	7.0	3.2	4.7	1.4	
100	6.3	3.3	6.0	2.5	

	species	species_name
1	0	setosa
50	1	versicolor
100	2	virginica

```
In [15]: # Converting the numbers to species names
iris_df['species_name'] = iris_df['species'].apply(lambda x: iris.target_names[x])
```

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First 3 classes of the dataset:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
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100	6.3	3.3	6.0	2.5	

	species	species_name
1	0	setosa
50	1	versicolor
100	2	virginica

```
In [20]: # Checking the dataset information
print("Dataset Information:")
iris_df.info()
```

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 150 entries, 0 to 149

Data columns (total 6 columns):

#	Column	Non-Null Count	Dtype
0	sepal length (cm)	150 non-null	float64
1	sepal width (cm)	150 non-null	float64
2	petal length (cm)	150 non-null	float64
3	petal width (cm)	150 non-null	float64
4	species	150 non-null	int64
5	species_name	150 non-null	str

dtypes: float64(4), int64(1), str(1)

memory usage: 7.2 KB

```
In [21]: # Describing the dataset
```

```
print("Statistical Summary:")
print(iris_df.describe())
```

Statistical Summary:

	sepal length (cm)	sepal width (cm)	petal length (cm)	\
count	150.000000	150.000000	150.000000	
mean	5.843333	3.057333	3.758000	
std	0.828066	0.435866	1.765298	
min	4.300000	2.000000	1.000000	
25%	5.100000	2.800000	1.600000	
50%	5.800000	3.000000	4.350000	
75%	6.400000	3.300000	5.100000	
max	7.900000	4.400000	6.900000	

	petal width (cm)	species
count	150.000000	150.000000
mean	1.199333	1.000000
std	0.762238	0.819232
min	0.100000	0.000000
25%	0.300000	0.000000
50%	1.300000	1.000000
75%	1.800000	2.000000
max	2.500000	2.000000

```
In [22]: # Counting the category
print("Species Counts:")
print(iris_df["species"].value_counts())
```

```
Species Counts:
species
0      50
1      50
2      50
Name: count, dtype: int64
```

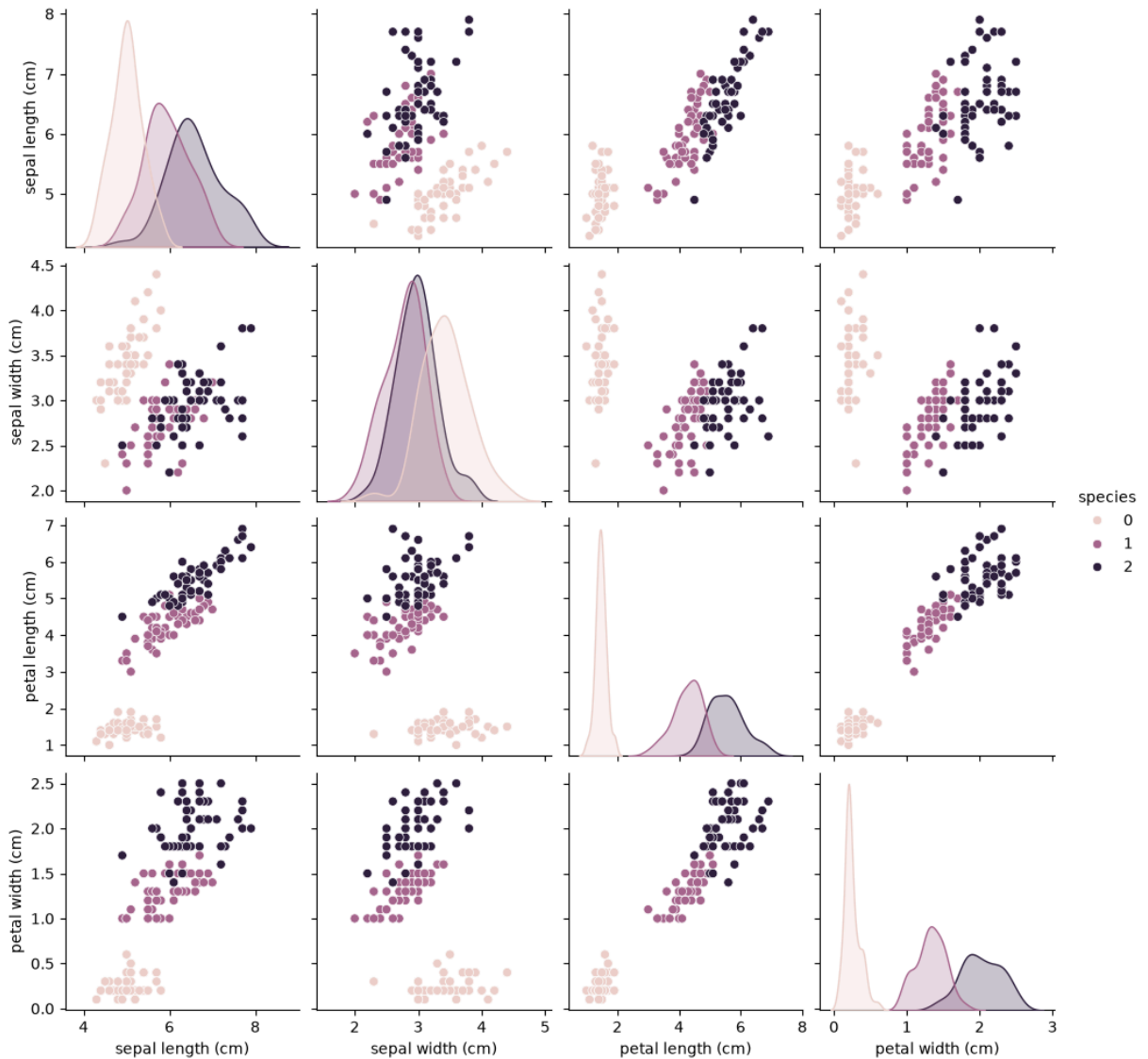
```
In [23]: # For normal dataset, we should now all the null values
print("Missing Values:")
print(iris_df.isnull().sum())
```

```
Missing Values:
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
species              0
species_name         0
dtype: int64
```

There are no missing values in the Iris dataset, so no missing-value treatment is required before generating the pairplot.

```
In [10]: # 7. Creating coloured pairplot
sns.pairplot(
    iris_df,
    hue="species"
)

plt.show()
```



Results and Interpretation

The coloured pairplot shows the relationships between the four numerical features of the Iris dataset: sepal length, sepal width, petal length, and petal width.

The `hue="species"` parameter assigns different colours to Setosa, Versicolor, and Virginica.

Observations

1. Petal length and petal width

These features show clear separation between the three species. Setosa forms a distinct cluster, while Versicolor and Virginica are relatively closer to each other.

2. Sepal length and sepal width

There is more overlap between the species compared with petal-related features.

3. Category separation

The coloured points help us identify how different species are distributed in feature space. The clearer the separation between colours, the easier it is to distinguish the categories using those features.

4. Overall observation

Petal length and petal width appear to be the most useful features for visually distinguishing the three Iris species.

Conclusion

The coloured pairplot provides an easy way to visualize relationships between multiple numerical features while distinguishing categories using different colours. In the Iris dataset, the species show better separation using petal length and petal width than using the sepal features. Therefore, the pairplot is useful for understanding patterns, relationships, and category separation in a dataset.

Declaration

I hereby declare that the work submitted in this practical assignment is my own and has not been copied from any other source.

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Date	11/07/2026

Prof. Prakash Ukhalkar

Course Teacher - Advanced Data Science Lab [MCA33PE21]

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